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$$16^x = 6 \cdot 4^x + 16 \Leftrightarrow (4^x)^2 = 6 \cdot 4^x + 16$$

$$4^x = t \Rightarrow t^2 = 6t + 16 \Leftrightarrow t^2 - 6t - 16 = 0$$

$$D = 36 - 4 \cdot 1 \cdot (-16) = 100$$

$$t_1 = \frac{6+10}{2} = 8$$

$$t_2 = \frac{6-10}{2} = -2$$

$$4^x = t_1 \Rightarrow 8 = 4^x \Leftrightarrow 2^3 = 2^{2x} \Rightarrow 2x = 3 \Leftrightarrow x = \frac{3}{2}$$

References